

Candidate Name

Centre Number

Candidate Number



For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

PHYSICS

PAPER 4 Practical Test

6032/4

2 hours 30 minutes

SPECIMEN PAPER

Candidates answer on the question paper.

Additional materials:

As listed in Instructions to Supervisors

Scientific calculator

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.

Write your answers in the spaces provided on the question paper.

In Questions 1 and 2, you are expected to record all your observations as soon as these observations are made, and to plan the presentation of the records so that it is not necessary to make a fair copy of them. The working of the answers is to be handed in. Marks are mainly given for a clear record of the observations actually made, for their suitability and accuracy, and for the use made of them.

Routine precautions and theory are **not** wanted in Questions 1 and 2. You should, however, record any **special** precautions you have taken so as to aid accuracy. At the end of the examination, fasten any separate answer paper used securely to the question paper.

INFORMATION FOR CANDIDATES

Questions 1 and 2 carry 20 marks each and question 3 carries 10 marks.

Additional paper and graphs should be submitted **only** if it becomes **necessary** to do so.

You are advised to spend approximately one hour on each of Questions 1 and 2 and 30 minutes on Question 3.

You are reminded of the need for good English and clear presentation in your answers.

FOR EXAMINER'S USE	
1	
2	
3	
TOTAL	

This specimen paper consists of 9 printed pages and 2 lined pages.

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- 1 In this experiment, you will investigate how the beads, n , in a boiling tube displace the volume of the liquid displaced, ΔV .

(a) (i) Set up the apparatus as shown in Fig.1.1.

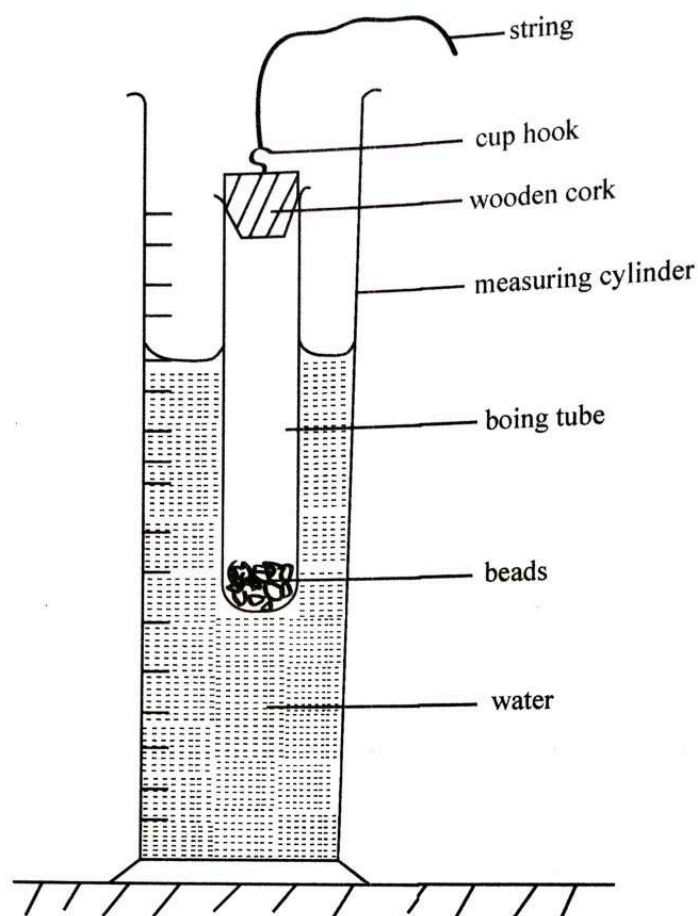


Fig.1.1

- (ii) Place the boiling tube into the measuring cylinder. Measure and record the initial volume, V' .

$$V' \text{ _____ } \pm \text{ _____ cm}^3.$$

- (iii) Place **20** beads into the boiling tube. Measure and record n and volume, V in your table of results. Include values of $\ln n$, $\ln(V-V')$ and absolute errors in $\ln(V-V')$.
- (iv) Repeat (iii) for n in the range $20 \leq n \leq 120$ to obtain **five** further observations.

- (b) It is suggested that n and $\Delta V = (V - V')$ are related by the equation.

$$n = \beta(\Delta V)^\alpha.$$

- (i) Plot the graph of $\ln n$ against $\ln(\Delta V)$ including error bars.
- (ii) Determine the gradient of the graph and use it to find the value of α with its uncertainty.

- (iii) Determine the intercept of the graph and use it to determine a value of β .

Measurements and calculations

For
Examiner's
Use
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**For
Examiner's
Use**

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2

In this experiment you will investigate the relationship between output voltage, V_0 , of an operational amplifier and the length, l , of the section of a potential divider wire.

- (a) (i) Complete the circuit shown in Fig. 2.1.

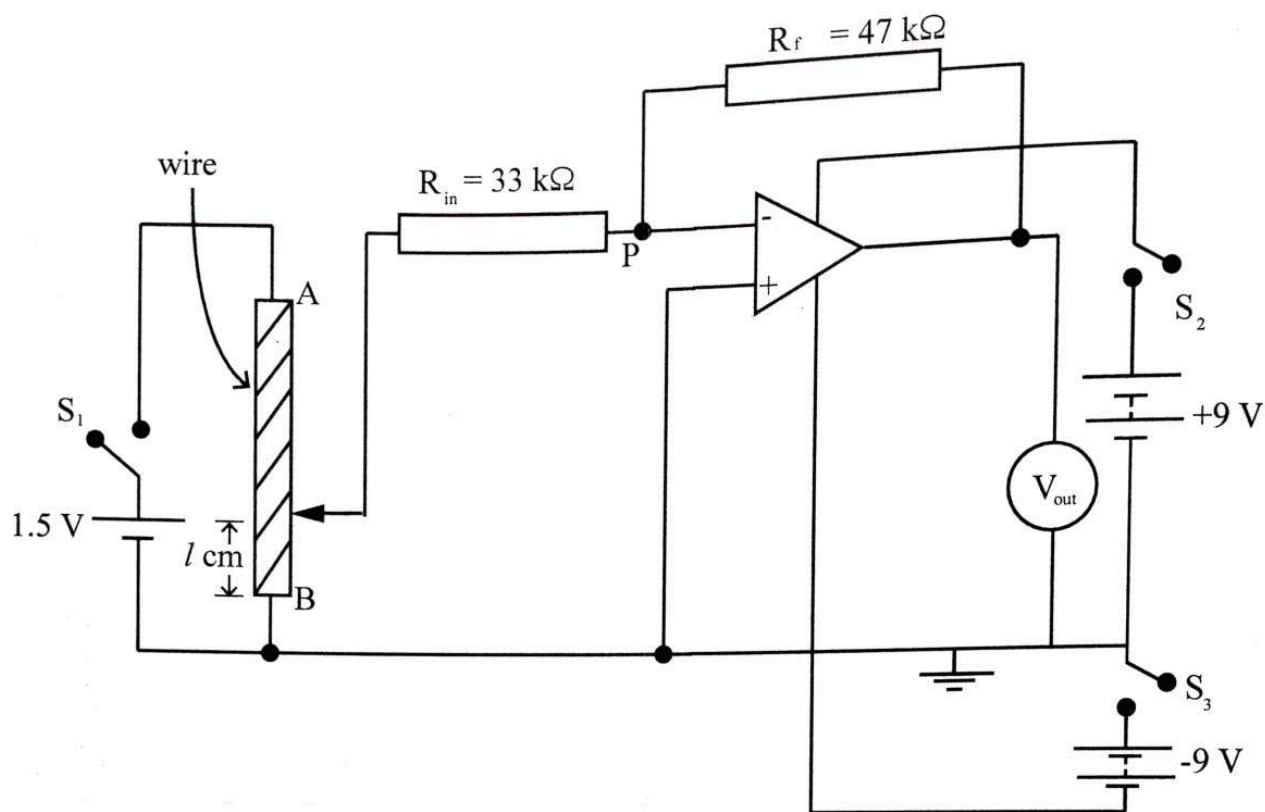


Fig. 2.1

- (ii) Close S_1 , S_2 and S_3 , measure and record V_0 when l is approximately 20.0 cm in a table of results.
- (iii) Repeat (a)(ii) by adjusting l in the range $20.0 \text{ cm} \leq l \leq 80.0 \text{ cm}$ to obtain five further observations of l and V_0 .
- (b) It is suggested that $V_0 = kl$ where k is a gain dependent constant.
- (i) Plot a graph of V_0 against l .
- (ii) Determine the gradient, hence the value of k .
- (iii) Determine the uncertainty in the value of k using the graph.
- (iv) Justify the number of significant figures you have quoted for k .
- (v) Suggest any **one** improvement to the experiment to improve its accuracy.

Measurements and calculation**For
Examiner's
Use**

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